

Fourier Transform of Sinusoids?

$$e^{j\Omega_0 t} \xleftarrow{FT} 2\pi\delta(\Omega - \Omega_0)$$

For fixed frequency Ω_0 , we can synthesize $e^{j\Omega_0 t}$ as a linear combination of complex sinusoids, that is,

$$e^{j\Omega_0 t} = 1 \cdot e^{j\Omega_0 t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\Omega - \Omega_0) e^{j\Omega t} d\Omega.$$

In other words, the synthesis direction is okay.

How about the other direction?

$$e^{j\Omega_0 t} \xrightarrow{FT} 2\pi\delta(\Omega - \Omega_0)???$$

$$1 \longleftrightarrow 2\pi \delta(\Omega)$$

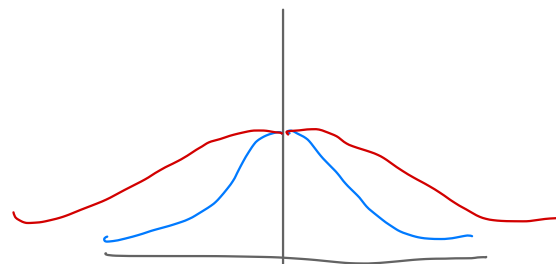
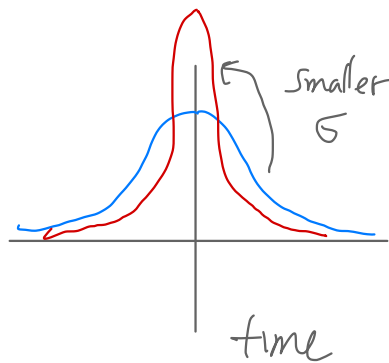
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modulation or freq-shift property

Fourier Transform as a Limit

$$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{FT} e^{-\frac{\sigma^2\Omega^2}{2}}$$

∫ is 1 *peak is 1*



smaller σ makes wider with peak fixed to 1

$$\lim_{\sigma \rightarrow 0} \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{FT} \lim_{\sigma \rightarrow 0} e^{-\frac{\sigma^2\Omega^2}{2}}$$

$\delta(t)$ 1

$\delta(t)$

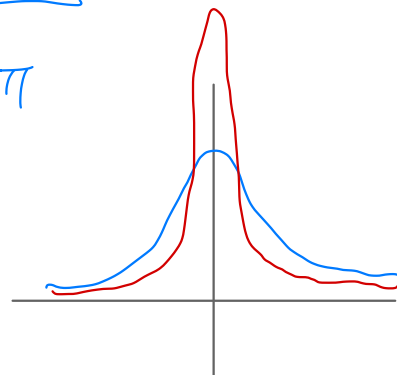
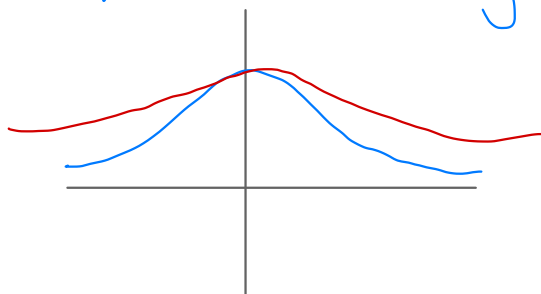
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$$\int \delta(t) e^{-j\omega t} dt = 1 \quad \text{sifting}$$

$$e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{FT} 2\pi \cdot \frac{\sigma}{\sqrt{2\pi}} e^{-\frac{\sigma^2\Omega^2}{2}}$$

peak = 1

$$\int = 2\pi$$



$$\lim_{\sigma \rightarrow \infty} e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{FT} \lim_{\sigma \rightarrow \infty} 2\pi \cdot \frac{\sigma}{\sqrt{2\pi}} e^{-\frac{\sigma^2\Omega^2}{2}}$$

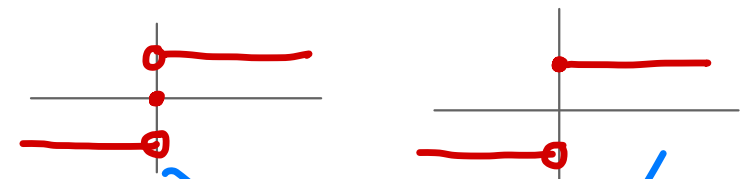
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$2\pi \delta(\Omega)$

$$\int_{-\infty}^{\infty} 1 \cdot e^{-j\Omega t} dt$$

is not defined

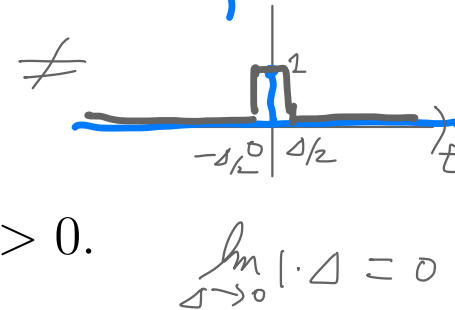
Fourier Transform of Sign Function



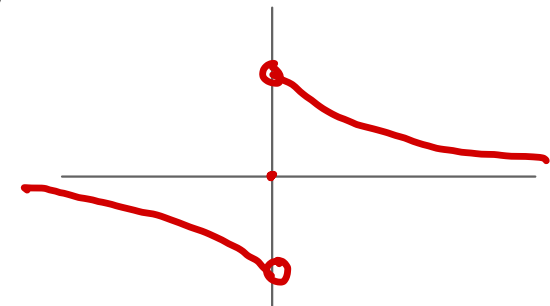
$$\text{sgn}(t) \xleftrightarrow{FT} \frac{2}{j\Omega}$$

Proof:

$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases} \quad x_a(t) = \begin{cases} e^{-at} & t > 0 \\ 0 & t = 0 \\ -e^{at} & t < 0 \end{cases} \quad a > 0.$$



$$x_a(t) = e^{-at}u(t) + -e^{at}u(-t)$$



$$e^{-at}u(t) \xleftrightarrow{FT} \frac{1}{a + j\Omega}, \quad e^{at}u(-t) \xleftrightarrow{FT} \frac{1}{a - j\Omega}$$

$$\text{sgn}(t) = \lim_{a \rightarrow 0} x_a(t) \xleftrightarrow{FT} \lim_{a \rightarrow 0} \frac{1}{a + j\Omega} - \frac{1}{a - j\Omega}$$

Fourier Transform of Unit Step Function

$$u(t) \xleftrightarrow{FT} \frac{1}{j\Omega} + \pi\delta(\Omega)$$

Proof:

$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}, \quad u(t) = \begin{cases} 1 & t > 0 \\ 1/2 & t = 0 \\ 0 & t < 0 \end{cases}$$

$$u(t) = \frac{\text{sgn}(t) + 1}{2}$$

$$\text{sgn}(t) \xleftrightarrow{FT} \frac{2}{j\Omega}, \quad 1 \xleftrightarrow{FT} 2\pi\delta(\Omega)$$

Integration Property

$$X_c(j0) = \int_{-\infty}^{\infty} x(t) \underbrace{e^{-j0t}}_1 dt$$

$$\int_{-\infty}^t x(\lambda) d\lambda \xleftrightarrow{FT} \frac{X_c(j\Omega)}{j\Omega} + \pi X_c(j0) \delta(\Omega)$$

Proof: Note

$$\int_{-\infty}^t x(\lambda) d\lambda = \int_{-\infty}^{\infty} x(\lambda) \underbrace{u(t-\lambda)}_{\text{red underline}} d\lambda = x(t) * u(t).$$

Convolution property implies

$$= \int_0^t 1 \quad \begin{matrix} t-\lambda \geq 0 & \lambda \leq t \\ t-\lambda < 0 & \lambda > t \end{matrix}$$

$$x(t) * u(t) \xleftrightarrow{FT} X_c(j\Omega) U(j\Omega) = X_c(j\Omega) \left(\frac{1}{j\Omega} + \pi \delta(\Omega) \right)$$

$$= \frac{X_c(j\Omega)}{j\Omega} + \pi X_c(j\Omega) \delta(\Omega)$$

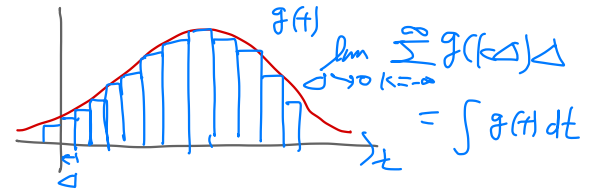
$$= \frac{X_c(j\Omega)}{j\Omega} + \pi X_c(j0) \delta(\Omega)$$

Assume $\int_{-\infty}^{\infty} x(t) dt = 0$

Let $y(t) = x(t) + C$

$$\int_{-\infty}^t y(t) dt = \underbrace{\int_{-\infty}^t x(t) dt}_{\text{blue underline}} + \int_{-\infty}^t C dt$$

Riemann
integral



Synthesis
FS representation

- Using FS, we can relate $X_c^{(P)}(j\Omega)$ to $x^{(P)}(t)$ as follows:

$$x^{(P)}(t) = \sum_{k=-\infty}^{\infty} x_k^{(P)} e^{j\frac{2\pi}{P}tk} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X_c^{(P)}(j\Omega) e^{j\Omega t} \frac{2\pi}{P} \Big|_{\Omega = \frac{2\pi k}{P}}$$

- Key idea: as P increases, the # of spectral lines per Hz in $X_c^{(P)}(j\Omega)$ increases, so that $X_c^{(P)}(j\Omega)$ approaches a continuum! In this case, Riemann sum arguments allow us to write the infinite sum as an integral. So, for $x(t) \triangleq \lim_{P \rightarrow \infty} x^{(P)}(t)$, which is no longer periodic, and $X_c(j\Omega) \triangleq \lim_{P \rightarrow \infty} X_c^{(P)}(j\Omega)$, we get

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j\Omega t} d\Omega$$

$$P X_c^{(P)} \longrightarrow X_c(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt, \quad \Big|_{\Omega = \frac{2\pi k}{P}}$$

$e^{j\Omega t} \cdot x(t)$

known as the CTFT.

- The convolution operation $y(t) = \int h(\tau) x(t - \tau) d\tau$ becomes algebraic under the CTFT:

$$\begin{aligned} Y_c(j\Omega) &= \int \left[\int h(\tau) x(t - \tau) d\tau \right] e^{-j\Omega t} dt \\ &= \int \int h(\tau) e^{-j\Omega \tau} x(t - \tau) e^{-j\Omega(t - \tau)} dt d\tau \\ &= \int h(\tau) e^{-j\Omega \tau} d\tau \int x(\rho) e^{-j\Omega \rho} d\rho \\ &= H_c(j\Omega) X_c(j\Omega). \end{aligned}$$

The CTFT — important transform pairs:

- The CTFT: $X_c(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt$ for $\Omega \in \mathbb{R}$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega)e^{j\Omega t} d\Omega,$$

is often written in “transform pair” notation

$$x(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega).$$

- Some important examples of CTFT transform pairs are:

- Dirac delta:

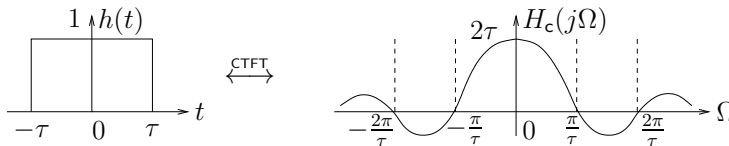
$$\delta(t - \tau) \xleftrightarrow{\text{CTFT}} e^{-j\Omega\tau}$$

- Complex exponential:

$$e^{j\Omega_0 t} \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega - \Omega_0)$$

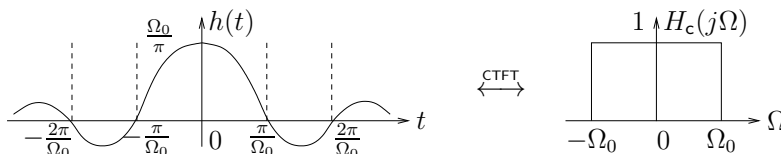
- Rectangle:

$$\begin{cases} 1 & \text{if } |t| < \tau \\ 0 & \text{else} \end{cases} \xleftrightarrow{\text{CTFT}} 2\frac{\sin(\Omega\tau)}{\Omega}$$



- Sinc:

$$\frac{\sin(\Omega_0 t)}{\pi t} \xleftrightarrow{\text{CTFT}} \begin{cases} 1 & \text{if } |\Omega| < \Omega_0 \\ 0 & \text{else} \end{cases}$$



The CTFT — derivations:

- Gaussian pulse

$$\frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{\text{CTFT}} e^{-\frac{\sigma^2\Omega^2}{2}}$$

- Impulse

$$\lim_{\sigma \rightarrow 0} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{\text{CTFT}} \lim_{\sigma \rightarrow 0} e^{-\frac{\sigma^2\Omega^2}{2}}$$

- Constant

$$\lim_{\sigma \rightarrow \infty} e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{\text{CTFT}} \lim_{\sigma \rightarrow \infty} 2\pi \cdot \frac{\sigma}{\sqrt{2\pi}} \cdot e^{-\frac{\sigma^2\Omega^2}{2}}$$

The CTFT — important properties:

Below, assume $x(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega)$ and $w(t) \xleftrightarrow{\text{CTFT}} W_c(j\Omega)$.

1. Linearity:

$$\alpha x(t) + \beta w(t) \xleftrightarrow{\text{CTFT}} \alpha X_c(j\Omega) + \beta W_c(j\Omega)$$

2. Conjugate symmetry:

$$x^*(t) = x(-t) \Leftrightarrow X_c(j\Omega) \in \mathbb{R}$$

$$x(t) \in \mathbb{R} \Leftrightarrow X_c^*(j\Omega) = X_c(-j\Omega)$$

3. Shift/modulation:

$$x(t - \tau) \xleftrightarrow{\text{CTFT}} e^{-j\Omega\tau} X_c(j\Omega)$$

$$e^{j\Omega_0 t} x(t) \xleftrightarrow{\text{CTFT}} X_c(j(\Omega - \Omega_0))$$

4. Reversal:

$$x(-t) \xleftrightarrow{\text{CTFT}} X_c(-j\Omega)$$

5. Multiplication/convolution:

$$x(t)w(t) \xleftrightarrow{\text{CTFT}} \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\theta) W_c(j(\Omega - \theta)) d\theta$$

$$\int_{-\infty}^{\infty} x(\tau) w(t - \tau) d\tau \xleftrightarrow{\text{CTFT}} X_c(j\Omega) W_c(j\Omega)$$

6. Energy conservation (“Parseval's theorem”):

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_c(j\Omega)|^2 d\Omega$$

The CTFT — integration property:

- CTFT of sign function

$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases} \xleftrightarrow{\text{CTFT}} \frac{2}{j\Omega}$$

Proof: Let $x_a(t) = e^{-at}u(t) - e^{at}u(-t)$ for $a > 0$. Since

$$e^{-at}u(t) \xleftrightarrow{\text{CTFT}} \frac{1}{a + j\Omega}, \quad e^{at}u(-t) \xleftrightarrow{\text{CTFT}} \frac{1}{a - j\Omega},$$

we have

$$\text{sgn}(t) = \lim_{a \rightarrow 0} x_a(t) \xleftrightarrow{\text{CTFT}} \lim_{a \rightarrow 0} \left(\frac{1}{a + j\Omega} - \frac{1}{a - j\Omega} \right).$$

- CTFT of step function

$$u(t) \xleftrightarrow{\text{CTFT}} \frac{1}{j\Omega} + \pi\delta(\Omega)$$

Proof: Note that $u(t) = \frac{\text{sgn}(t) + 1}{2}$. Recall

$$\text{sgn}(t) \xleftrightarrow{\text{CTFT}} \frac{2}{j\Omega}, \quad 1 \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega).$$

- Integration property

$$\int_{-\infty}^t x(\lambda) d\lambda \xleftrightarrow{\text{CTFT}} \frac{X_c(j\Omega)}{j\Omega} + \pi X_c(j0) \delta(\Omega)$$

Proof: Note

$$\int_{-\infty}^t x(\lambda) d\lambda = \int_{-\infty}^{\infty} x(\lambda) u(t - \lambda) d\lambda = x(t) * u(t).$$

Due to the convolution property, we have

$$x(t) * u(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega) \left(\frac{1}{j\Omega} + \pi \delta(\Omega) \right).$$

$$x(t) = e^{j\Omega_0 t} \quad \text{for some } \Omega_0$$

$$Y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\lambda) x(t-\lambda) d\lambda$$

$$= \int_{-\infty}^{\infty} h(\lambda) e^{j\Omega_0(t-\lambda)} d\lambda$$

$$= \underbrace{e^{j\Omega_0 t}}_{x(t)} \int_{-\infty}^{\infty} h(\lambda) e^{-j\Omega_0 \lambda} d\lambda$$

$$\frac{Y(t)}{x(t)} = \underbrace{\int_{-\infty}^{\infty} h(\lambda) e^{-j\Omega_0 \lambda} d\lambda}$$

When this quantity is finite, it does not depend on time t .

$\Rightarrow$ We conclude $e^{j\Omega_0 t}$ is an eigenfunction of the system.

example

$$h(t) = u(t)$$

$$Y(t) = h(t) * x(t) = \int_{-\infty}^t x(\lambda) d\lambda$$

$u(t)$ - derivative

$$\int h(t) e^{-j\Omega t} dt = \frac{1}{j\Omega} + \pi \delta(\Omega)$$

$$X(t) = e^{st}$$

$$s \in \mathbb{C}$$

$$Y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\lambda) e^{s(t-\lambda)} d\lambda$$

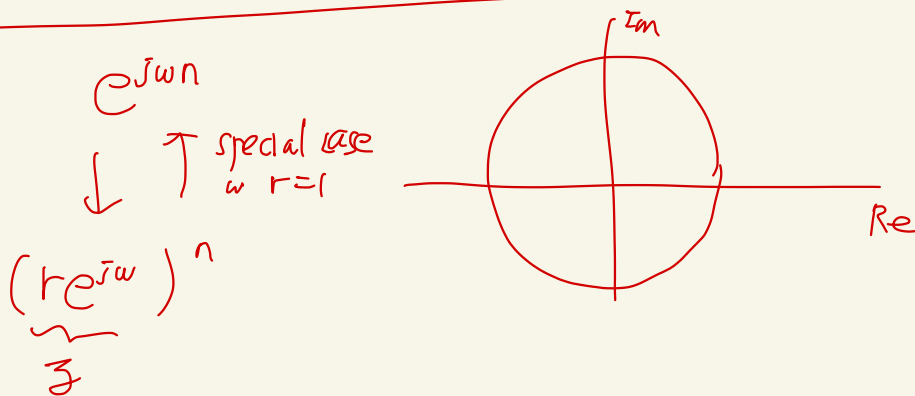
$$= e^{st} \underbrace{\int_{-\infty}^{\infty} h(\lambda) e^{-s\lambda} d\lambda}$$

If this is finite, then

e^{st} is an eigenfunction of the system with $h(t)$ LTI

$$s = \sigma + j\omega \quad \sigma, \omega \in \mathbb{R}$$

$$e^{st} = e^{(\sigma + j\omega)t} = \underbrace{e^{\sigma t}}_{\text{exponential}} \cdot \underbrace{e^{j\omega t}}_{\text{complex sinusoid}}$$



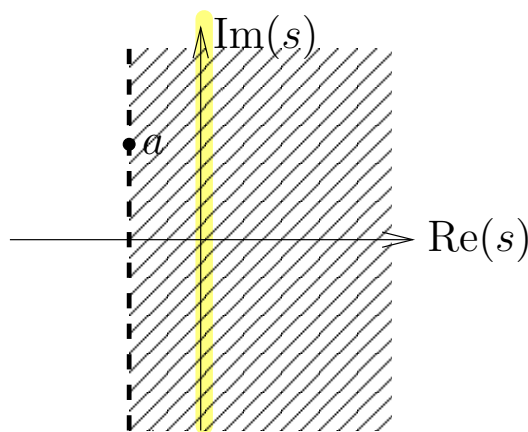
The Laplace transform:

- Recall the 2-sided *Laplace transform* (LT) of $x(t)$:

$$X_c(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt, \text{ for } s \in \mathbb{C} \text{ such that } |X_c(s)| < \infty.$$

- The CTFT can be interpreted as a special case of the LT for which $s = j\Omega$ with $\Omega \in \mathbb{R}$, and that is why we write the CTFT of $x(t)$ as “ $X_c(j\Omega)$.”
- The LT of a given $x(t)$ is defined only for those $s \in \mathbb{C}$ such that $|X_c(s)| < \infty$. This portion of the complex plane is known as the “*region of convergence (ROC) of $X_c(s)$* .”

For example, if $x(t) = e^{at}u(t)$, where $a \in \mathbb{C}$ and $u(t)$ is the Heaviside step function ($u(t) = 1$ for $t \geq 0$ and $u(t) = 0$ otherwise), then one can show that the ROC is the set $\{s \in \mathbb{C} : \text{Re}(s) > \text{Re}(a)\}$.



- Like the CTFT, the utility of the LT stems from the fact that the convolution integral $y(t) = \int h(\tau)x(t - \tau)dt$ simplifies to the algebraic expression $Y_c(s) = H_c(s)X_c(s)$.
- For many signals $x(t)$ (e.g., impulse, step, ramp, decaying exponentials), there exist convenient closed-form expressions for $X_c(s)$. Moreover, when $Y_c(s)$ is a ratio of polynomials, its inverse LT can be calculated using *partial fraction expansion* (PFE) and table lookup.
- The *advantage* of the LT vs. the CTFT is that there is a much larger class of signals with closed-form transforms. The *disadvantage* of the LT vs. the CTFT is that the inverse transform step (requiring PFE or contour integration) is more complicated.

In practical DSP, the FT is computed numerically and so we don't care too much about closed-form transforms.

Thus, in this course, we focus on the simpler CTFT.

Stability of LTI systems:

An LTI system is stable if its impulse response $h(t)$ satisfies one of the following conditions.

- Time-domain viewpoint:

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \quad (h(t) \text{ is absolutely integrable})$$

- Fourier transform viewpoint:

$$\left| \int_{-\infty}^{\infty} h(t) e^{-j\Omega t} dt \right| < \infty, \quad \forall \Omega \quad (\text{FT exists})$$

- Laplace transform viewpoint:

For causal rational $h(t)$, all poles of $H(s)$ are located in open left-half plane

DT-LTI systems:

- So far, we have reviewed continuous-time LTI systems:
 1. Eigenfunctions and FS (covers periodic signals)
 2. CTFT (extension to non-periodic signals)
 3. LT (“deluxe” version of the CTFT)
- Now, we discuss some counterparts for discrete-time LTI systems:
 - a. DTFT (analogous to CTFT)
 - b. Z-transform (analogous to LT)

We postpone our discussion of the DFT, the discrete-time counterpart to FS, until a bit later.

$$e^{j(\omega+2\pi)n} = e^{j\omega n + j2\pi n} = e^{j\omega n} e^{j2\pi n}$$

The DTFT:

- The *discrete-time Fourier transform* (DTFT) is defined as

analysis

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \text{ for } \omega \in \mathbb{R}$$

dot product of $e^{j\omega n}$ and $x[n]$

synthesis

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega,$$

where the notation “ $X(e^{j\omega})$ ” is consistent with the Z-transform. (More on this later...)

The lack of the subscript “ c ” distinguishes the DTFT and Z-transform “ $X(\cdot)$ ” from the CTFT and LT “ $X_c(\cdot)$ ”.

- Essentially, the DTFT gives a “continuous-frequency” characterization of DT signals in the same way that the CTFT gave a “continuous-frequency” characterization of CT signals. (The DTFT/CTFT relationship will be explained further in the context of *sampling*.)
- Important: the DTFT is 2π -periodic in ω :

$$X(e^{j\omega}) = X(e^{j(\omega+2\pi k)}) \text{ for any integer } k,$$

because the function $e^{j\omega}$ is itself 2π -periodic in ω .

In contrast, the CTFT $X_c(j\Omega)$ is not periodic in Ω !